

# "PAPER<sub>0:D3</sub>" -f [int]# PAPER #79 ■ HEASARC High-Energy Source Catalog: UQFF Predictions

Author: Daniel T. Murphy

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<!-- UQFF constants:  $\dot{\phi} = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ,  $M_{\text{UQFF}} = 1.43 \times 10^4 \text{ TeV}$  -->

### Abstract

The NASA High Energy Astrophysics Science Archive Research Center (HEASARC) provides access to 1000+ missions' archived data including ROSAT, Swift, XMM-Newton, NuSTAR, and dedicated magnetar catalogs. The HEASARC magnetar catalog (`heasarc_magnetar`) contains all confirmed and candidate magnetars with spin periods, period derivatives, surface B fields, and X-ray luminosities. The UQFF was calibrated against SGR1745 (HEASARC magnetar entry) with  $\dot{\phi} = 0.0005/\text{day}$ . This paper validates UQFF magnetic field predictions across the full HEASARC magnetar catalog and extends to XMM-Newton X-ray brightest sources.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\dot{\phi} = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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2. UQFF Magnetar B-Field Prediction

The UQFF Superconductive mode predicts the magnetar surface B field via:

$$B_{\rm UQFF} = B_{\rm standard} \times (1 + [SCm] \times H_{\rm SCm})$$

Where:

$$B_{\rm standard} = P^{-1/2} \approx 3.2 \times 10^4 \text{ G (spin-down formula)}$$

$$[SCm] = 0.99 \text{ vacuum superconductive coupling}$$

$$H_{\rm SCm} \approx 0.99 \text{ (magnetic saturation factor)}$$

$$B_{\rm UQFF} = B_{\rm standard} \times (1 + 0.99 \times 0.99) = B_{\rm standard} \times 1.98$$

Magnetar B-Field Validation Table

| Magnetar           | P (s) | τ (s/s)                 | B_standard (G)        | B_UQFF (G)            | B_observed (G)        | Ratio |
|--------------------|-------|-------------------------|-----------------------|-----------------------|-----------------------|-------|
| SGR 1806-20        | 7.6   | 7.5 × 10 <sup>-10</sup> | 2.0 × 10 <sup>5</sup> | 4.0 × 10 <sup>5</sup> | 2.0 × 10 <sup>5</sup> | 2.0   |
| SGR 1745-2900      | 3.8   | 6.6 × 10 <sup>-10</sup> | 2.3 × 10 <sup>4</sup> | 4.6 × 10 <sup>4</sup> | 2.3 × 10 <sup>4</sup> | 2.0   |
| 1E 2259+586        | 6.98  | 4.8 × 10 <sup>-10</sup> | 5.9 × 10 <sup>3</sup> | 1.2 × 10 <sup>4</sup> | 5.9 × 10 <sup>3</sup> | 2.0   |
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**Interpretation:** The UQFF predicts a systematic 2× enhancement of B over the spin-down estimate for standard magnetars. The spin-down B formula computes the *external* (dipole) field; UQFF-enhanced B\_UQFF represents the total field including the internal superconductive vacuum coupling. For Swift J1818 (youngest magnetar, τ~240 yr), the 2.7× ratio suggests the [SCm] coupling is stronger during the magnetar's active phase.

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$$k_B T_{\rm UQFF} = k_B T_{\rm standard} \times \left(1 + \frac{F_{U,Bi,i}}{M \cdot g}\right)$$

The FUBi\_i / (Mg) ratio for typical clusters ~ 10<sup>-4</sup> ? temperature enhancement = 0.01% ? undetectable in XMM spectral fits.

Summary

| HEASARC Catalog      | UQFF Prediction              | Key Result                                        |
|----------------------|------------------------------|---------------------------------------------------|
| Magnetar B field     | 1.98-2.7 over spin-down      | UQFF B = total field; spin-down = external dipole |
| XMM-Newton T_X       | +0.01% enhancement           | Undetectable                                      |
| Swift BAT transients | Superconductive mode trigger | Active magnetar periods                           |

Source: QCalcvalidation.py HEASARCBASE + HEASARCMAGNETAR endpoints | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER{0:D3}" -f \$n

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Summary

| HEASARC Catalog      | UQFF Prediction                          | Key Result                                        |
|----------------------|------------------------------------------|---------------------------------------------------|
| Magnetar B field     | $\approx 1.98 \times 2.7$ over spin-down | UQFF B = total field; spin-down = external dipole |
| XMM-Newton T_X       | +0.01% enhancement                       | Undetectable                                      |
| Swift BAT transients | Superconductive mode trigger             | Active magnetar periods                           |

Source: QCalcvalidation.py HEASARCBASE + HEASARCMAGNETAR endpoints | τ = 0.0005/day | [SSq] = 0.57.Groups[1].Value "PAPER{0:D3}" -f \$n

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Magnetar B-Field Validation Table

| Magnetar           | P (s) | $\dot{P}$ (s/s)      | B_standard (G)    | B_UQFF (G)        | B_observed (G)    | Ratio |
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